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ACADEMIC PROJECT REPORT: WINE QUALITY CLASSIFICATION
USING RANDOM FOREST CLASSIFIER
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Course: Machine Learning & Data Science
Task: Supervised Classification Pipeline & Interactive Deployment
Student Name: [Your Name / Student ID]
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1. PROBLEM STATEMENT

Sensory wine quality grading by human panels is inherently subjective, logistically cumbersome, and prone to inconsistency. The objective of this study is to develop an objective, automated Machine Learning classification model that accurately predicts the quality tier of Vinho Verde red wine based exclusively on 11 physicochemical laboratory measurements. Specifically, the model solves a binary classification task: distinguishing high-grade "Premium Wine" (sensory score >= 7 out of 10) from "Standard Wine" (sensory score < 7 out of 10), providing winemakers and distributors with an empirically validated quality assurance mechanism.

2. DATA PREPROCESSING STEPS

The benchmark dataset consists of 1,599 instances sourced from the UCI Machine Learning Repository (Cortez et al., 2009). The preprocessing pipeline executed the following sequential stages:

a. Data Hygiene & Integrity: Evaluated missing values; the dataset contained 0 null or unrecorded entries across all 11 attributes.
b. Target Binarization: Transformed the original discrete quality score (ranging from 3 to 8) into a binary target vector:
   - Class 1 (Premium): Quality >= 7 (217 samples, 13.57% of dataset)
   - Class 0 (Standard): Quality < 7 (1,382 samples, 86.43% of dataset)
c. Stratified Train-Test Splitting: Partitioned data into 80% training (1,279 samples) and 20% test (320 samples) partitions. Stratification was strictly enforced to preserve identical class ratios across both splits, preventing distribution skew during model evaluation.
d. Feature Matrix Alignment: Maintained all 11 physicochemical continuous features in fixed canonical ordering to ensure complete pipeline reproducibility.

3. ALGORITHM SELECTION: WHY RANDOM FOREST?

The Random Forest Classifier was chosen as the optimal architecture for this domain due to several foundational theoretical advantages:

a. Non-Linear Feature Interactions: Wine quality does not exhibit simple linear separability. Random Forest effectively captures complex non-linear thresholds (e.g., the interactive relationship between alcohol, acidity, and sulphates).
b. Robustness Against Overfitting: By constructing an ensemble of 100 decorrelated decision trees trained on bootstrapped subsets (bagging) with random feature subspaces at each node split, Random Forest significantly mitigates the variance and overfitting tendencies of individual decision trees.
c. Invariance to Feature Scaling: Unlike Support Vector Machines or Logistic Regression, tree-based partition boundaries are monotonic and do not require normalization or standardization, preserving the original physical interpretability of the units.
d. Built-in Feature Importance: Gini impurity reduction directly quantifies the relative contribution of each chemical factor to final wine grade predictions.

4. EXPERIMENTAL EVALUATION METRICS

The model was tested on 320 unseen test samples using standard classification metrics:

- Overall Accuracy : 94.38% (302 out of 320 correct predictions)
- ROC-AUC Score : 0.9554 (Outstanding discriminatory power)
- Macro Average F1-Score : 0.86
- Weighted Average F1 : 0.94

Confusion Matrix Breakdown:

- True Negatives (Standard classified as Standard) : 275
- False Positives (Standard classified as Premium) : 2 (0.7% False Alarm Rate)
- False Negatives (Premium classified as Standard) : 16
- True Positives (Premium classified as Premium)   : 27 (Precision: 93.1%)

Performance by Class:
- Class 0 (Standard): Precision = 0.95, Recall = 0.99, F1-Score = 0.97
- Class 1 (Premium) : Precision = 0.93, Recall = 0.63, F1-Score = 0.75

5. ANALYSIS OF RESULTS & CHEMICAL INSIGHTS

1. Exceptional Precision (93.1%): The model achieved an extraordinarily low false positive rate, generating only 2 false positive classifications. In commercial winemaking, this is critical: a bottle labeled as "Premium" has over a 93% certainty of truly being superior grade.
2. Key Chemical Drivers (Feature Importance Ranking):
   - Alcohol (17.37%): The single strongest predictor of high wine quality. Wines with alcohol >= 11.5% had dramatically higher probabilities of achieving >= 7 ratings.
   - Sulphates (11.13%): Potassium sulphate serves as an antimicrobial agent and antioxidant; higher concentrations correlate positively with structural preservation.
   - Volatile Acidity (10.24%): Acetic acid imparts a pungent vinegar taste. Premium wines uniformly maintained volatile acidity below 0.45 g/dm³.
   - Citric Acid (9.29%): Contributes natural freshness and tartness, enhancing sensory balance.
3. Interactive Deployment: The trained model was serialized (model.pkl) and deployed via Streamlit (app.py), providing real-time sliders for the top 4 predictors while maintaining empirical median baselines for the remaining 7 features.

END OF REPORT